

## Note on the Semantics of Frequency Adverbials and Mereotopology

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This is a brief note on Marcin Wągiel's SALT 33 2023 proceedings paper 'Acts, occasions, and multiplicatives: A mereotopological account' and the analysis and approach of frequency adverbials in *Parts and Wholes in Semantics* (OUP 1997).

Wągiel gives a formal semantic analysis of the event-internal reading (counting occasions) and the event-external reading (counting subevents) of (i), distinguishing two different syntactic positions for them:

(i) *John knocked three times at the door.*

The event-internal reading is formalized in terms of maximal temporal connectedness, the event-external reading in terms of a notion of event-integrated whole.

The same analysis Marcin proposed of 'times' was already in my 1997 OUP book *Parts and Wholes in Semantics* (Chap. 5). But according to Wągiel:

'In contrast, Moltmann (1997) emphasized the role of structured parthood in the domain of events, but her account entirely rejects mereology (for a critical review of her system see Pianesi 2002). For this reason I will pursue an entirely different approach based on the structured part-whole relations called mereotopology, which extends standard mereology with topological notions such as maximal connectedness.' (p. 227)

This is a misrepresentation and a misappropriation of the analysis in book *Parts and Wholes in Semantics*, OUP, 1997 <http://www.friederike-moltmann.com/uploads/Parts%20and%20Wholes%20in%20Semantics-s.pdf>

It also suggests a misappropriation of the overall view of *Parts and Whole in Semantics*.

That book introduced mereotopology into natural language semantics, against the background of extensional mereology (as adopted by Link and others), with a wide range of application, including the semantics of *many times* and other frequency expressions.

Here some further details about the actual analysis of *times* in *Parts and Wholes in Semantics* that should be of general interest:

The semantic analysis of 'times' and frequency adverbials in Chap. 5.1. (p.141ff) is a mereotopological account identical to the one Wągiel proposed for the first reading of (i). Or rather the account is one that makes use of a metrical space and then shows that that amounts to a mereotopological part structure based on temporal maximal connectedness. Pianesi in his 2002 review, rightly, points out that an account in terms of maximal connectedness would be simpler. But the answer to why a metrical structure is used is apparent from the next subsequent section (5.2.), which is about the German metrical quantifier 'manche' ('some'), which goes with frequency adverbials 'manchmal' 'sometimes': 'manche' imposes a relation of

distance (in some dimension) among individuals, and the latter need not involve maximal connectedness (but may be integrated wholes or single things in virtue of other conditions).

The notion of maximal connectedness (R-integrated whole), the central notion of mereotopology, is used throughout the book (and related papers) as the most important notion of integrated whole for the semantics of events, plurals and mass nouns.

*Parts and Wholes in Semantics* did not reject mereology; to the contrary it showed the importance of mereology (the theory of part-whole) for a huge range of semantic phenomena and introduced the notion of an integrated whole into semantic analysis (in addition to an ordering among parts) to account for them. What it did was to argue for a single part relation applying to different types of entities, in particular pluralities and quantities, and it gave a range of motivations for that. One of them comes from the semantics of frequency expressions themselves: 'frequent' applies to both plurals ('frequent rainfalls') and mass nouns ('frequent rain'), which requires a single part relation, or so was the argument (Section 5.1.).

There is in fact a third reading of 'times', on which 'times' count events or states at specific given occasions ('John was asleep three times') (a reading which in *Parts and Wholes in Semantics* is attributed to H. de Swart).

The three readings are also described here: ['Events in Contemporary Semantics'](#). In James Bahoh, Marta Cassina, and Sergio Genovesi (eds): *21st-Century Philosophy of Events: Beyond the Analytic / Continental Divide*. Edinburgh University Press, Edinburgh, 2025.

A remark concerning the Pianesi review, which is mentioned in the Wagiel quote:  
The review did not show that the 1997 approach failed, in relation to extensional mereology. In part it had pointed out weakness of analyses in the 1997 book of new generalizations that had not been (and to an extent still have not been) analysed elsewhere; in part it showed the general limits of mereology for the semantics of plurals (which made me pursue plural reference in my more recent work on plurals). The review rightly called the book 'programmatic' in its use of notions of integrated whole and maximal connectedness (entirely novel in semantics at the time), rather than being 'a self-contained essay'.